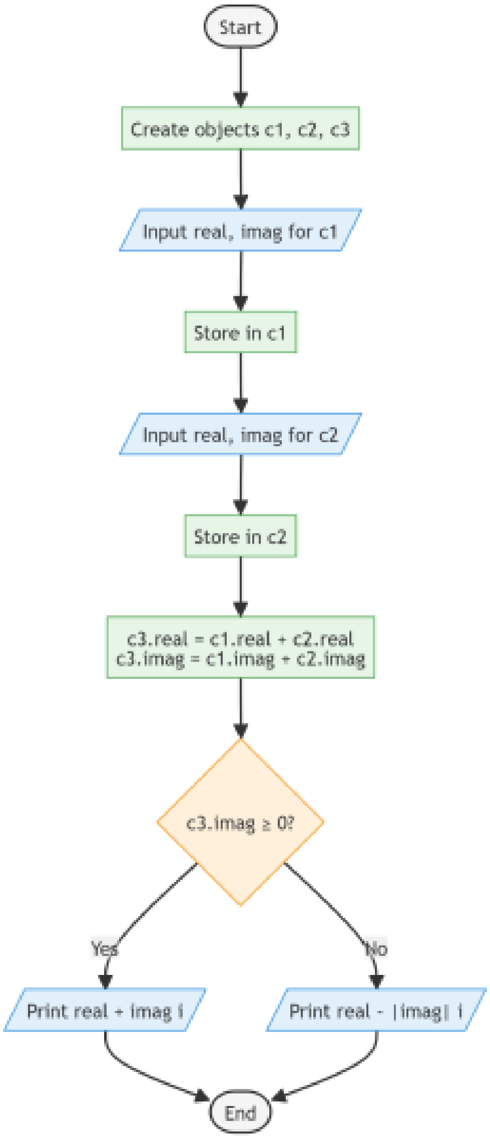
Experiment 14.1.1: Complex Number Addition using Class

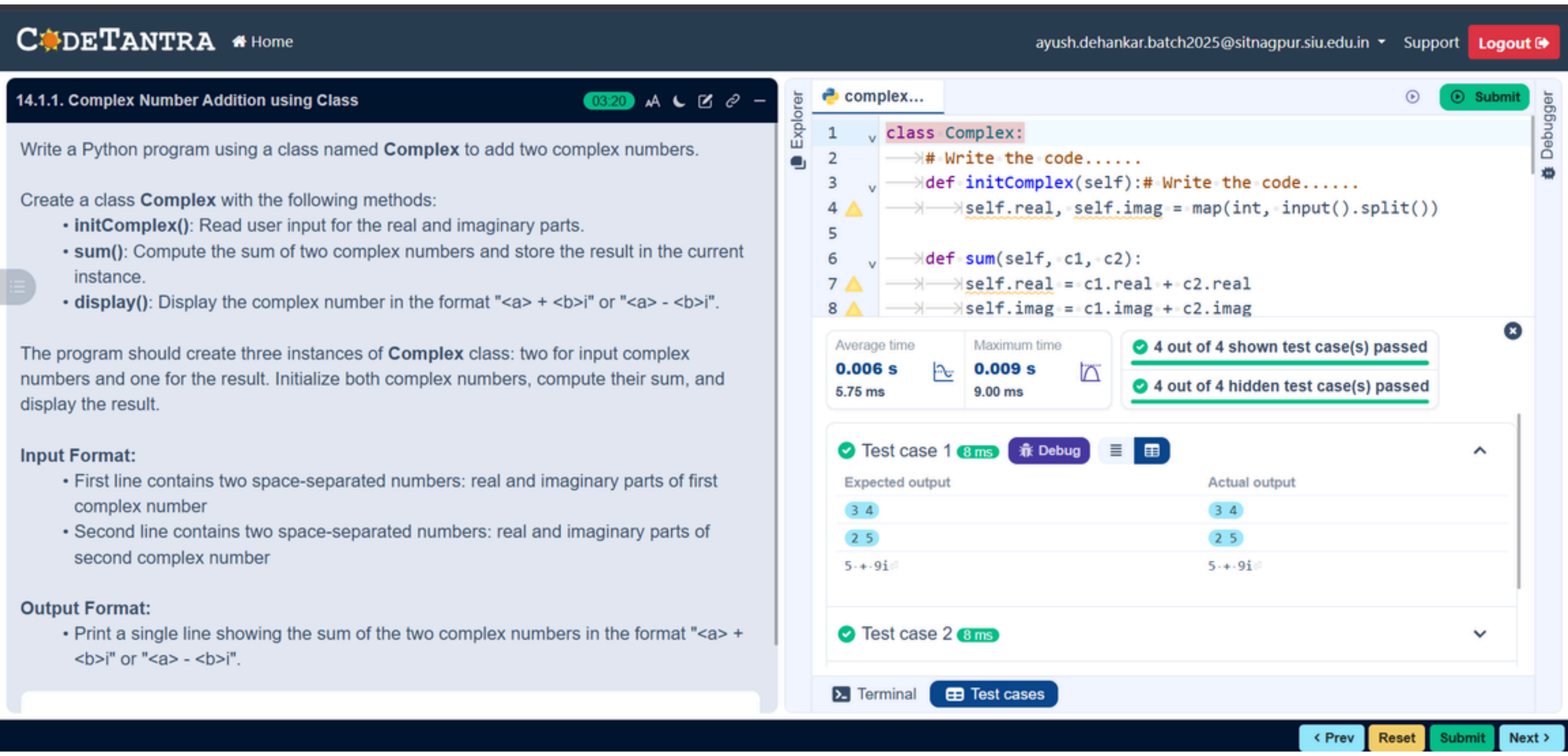
1. Aim

To design and implement a Python program using a class named **Complex** to add two complex numbers. The program involves using methods to initialize values from user input, compute the sum of two complex instances, and display the result in a standard complex number format ($a + bi$ or $a - bi$).

2. Pseudocode

1. **START**
2. **DEFINE** a class Complex:
   - **METHOD** initComplex(self):
     - **READ** two space-separated integers (real and imaginary parts) and **STORE** them in self.real and self.imag.
   - **METHOD** sum(self, c1, c2):
     - **SET** self.real = c1.real + c2.real.
     - **SET** self.imag = c1.imag + c2.imag.
   - **METHOD** display(self):
     - **IF** self.imag >= 0:
       - **PRINT** the complex number in the format: real + imag i.
     - **ELSE**:
       - **PRINT** the complex number in the format: real - abs(imag) i.
3. **MAIN PROGRAM**:
   - **CREATE** three instances of the Complex class: c1, c2, and c3.
   - **CALL** c1.initComplex() to read the first complex number.
   - **CALL** c2.initComplex() to read the second complex number.
   - **CALL** c3.sum(c1, c2) to compute the addition.
   - **CALL** c3.display() to output the result.
4. **END**


Start
Create objects c1, c2, c3
Input real, imag for c1
Store in c1
Input real, imag for c2
Store in c2
c3.real = c1.real + c2.real
c3.imag = c1.imag + c2.imag
c3.imag ≥ 0?
Yes
No
Print real + imag i
Print real - |imag| i
End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14.1.1. Complex Number Addition using Class
03:20
Write a Python program using a class named Complex to add two complex numbers.
Create a class Complex with the following methods:
• initComplex(): Read user input for the real and imaginary parts.
• sum(): Compute the sum of two complex numbers and store the result in the current instance.
• display(): Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".
The program should create three instances of Complex class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.
Input Format:
• First line contains two space-separated numbers: real and imaginary parts of first complex number
• Second line contains two space-separated numbers: real and imaginary parts of second complex number
Output Format:
• Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or "<a> - <b>i".
Explorer
complex...
Submit
Debugger
class Complex:
# Write the code......
def initComplex(self):# Write the code......
self.real, self.imag = map(int, input().split())
def sum(self, c1, c2):
self.real = c1.real + c2.real
self.imag = c1.imag + c2.imag
Average time
0.006 s
5.75 ms
Maximum time
0.009 s
9.00 ms
4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
8 ms
Debug
Expected output
Actual output
3 4
3 4
2 5
2 5
5 + 9i
5 + 9i
Test case 2
8 ms
Terminal
Test cases
Prev
Reset
Submit
Next